# AN AFFINE SCHEME WITH A G_M ACTION IS THE SPECTRUM OF A Z-GRADED RING

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

*Allowed standard prerequisites (editorial).*

- Example example-Gm-on-affine (canonical tag 0EKJ). cited at proof lines 2576-2578. Navigation for the standard expansion identity; the identity itself follows from the action axiom.

*Source reference.* Original section 0EKJ: sources/stacks-chow/tags.txt, source lines 18239–18239. Cited at proof lines 2576-2578 as the comparison computation; navigation for the standard identity, whose derivation from the action axiom is one line.

**Lemma 0.2.** *Let $a : \mathbf{G}_m \times X \to X$ be an action on an affine scheme. Then $X$ is the spectrum of a $\mathbf{Z}$-graded ring and the action is as in Example 0EKJ.*

*Proof.* Let $f \in A = \Gamma(X, \mathcal{O}_X)$. Then we can write

$$a^\sharp(f) = \sum\nolimits_{n \in \mathbf{Z}} f_n \otimes x^n \quad \text{in} \quad A \otimes \mathbf{Z}[x, x^{-1}] = \Gamma(\mathbf{G}_m \times X, \mathcal{O}_{\mathbf{G}_m \times X})$$

as a finite sum with $f_n$ in $A$ uniquely determined. Thus we obtain maps $A \to A$, $f \mapsto f_n$. Since $a$ is an action, if we evaluate at $x = 1$, we see $f = \sum f_n$. Since $a$ is an action we find that

$$\sum (f_n)_m \otimes x^m \otimes x^n = \sum f_n x^n \otimes x^n$$

(compare with computation in Example 0EKJ). Thus $(f_n)_m = 0$ if $n \neq m$ and $(f_n)_n = f_n$. Thus if we set

$$A_n = \{f \in A \mid f_n = f\}$$

then we get $A = \sum A_n$. On the other hand, the sum has to be direct since $f = 0$ implies $f_n = 0$ in the situation above. □

*Date*: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.